INCH-POUND
MIL-DTL-21097/5G
w/AMENDMENT 2
4 January 2022
SUPERSEDING
MIL-DTL-21097/5G
w/AMENDMENT 1
1 February 2017

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, PRINTED WIRING BOARD MALE ADAPTER, CONTACT SPACING (.200) ALTERNATE DUAL ROW, TYPE CS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-21097.

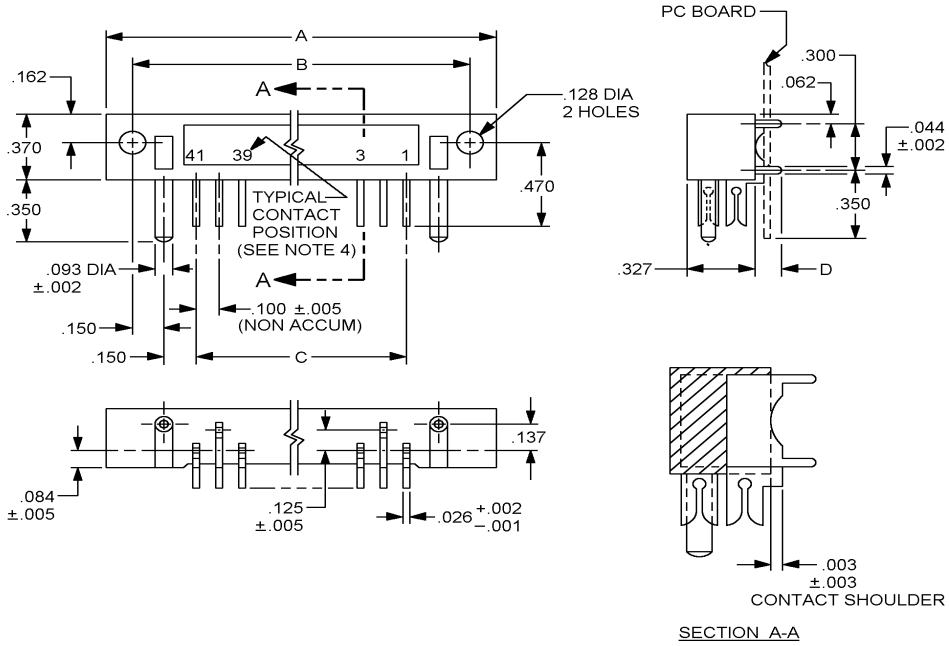


FIGURE 1. Alternate dual row contacts, type CS.

AMSC N/A
FSC 5935
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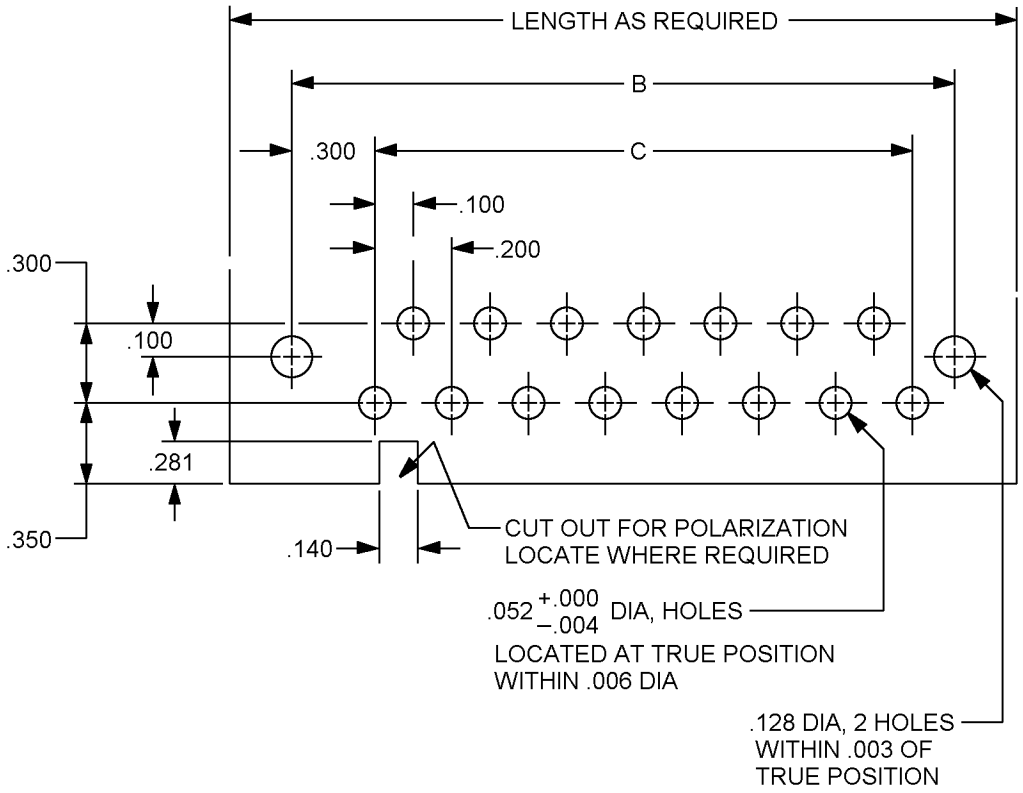
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Inches	mm	Inches	mm	Inches	mm
.001	.03	.084	2.13	.162	4.11
.002	.05	.093	2.36	.300	7.62
.003	.08	.100	2.54	.327	8.31
.005	.13	.125	3.18	.350	8.89
.026	.66	.128	3.25	.370	9.40
.044	1.12	.137	3.48	.470	11.94
.062	1.57	.150	3.81		

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ± .010 (.25 mm) for three place decimals.
3. Metric equivalents are given for information only.
4. Contacts shall be designated on the connector using numerals in the location shown.

FIGURE 1. Alternate dual row contacts, type CS. – Continued.

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Inches	mm
.003	.08
.004	.10
.006	.15
.052	1.32
.100	2.54
.128	3.25
.140	3.56
.200	5.08
.281	7.14
.300	7.62
.350	8.89

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ± .010 (.25 mm) for three place decimals.
3. Metric equivalents are given for information only.

FIGURE 2. Printed circuit board mounting and hole pattern.

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TABLE I. Connector, electrical, male adapter, type CS, class b, no mounting provision A.

Dash number	Style	Size	Dimensions (inches) 1/ 2/ A (maximum)	B ±0.015 (0.38)	C ±0.005 (0.13)	D ±0.010 (0.25)
01	17	2	2.470 (62.74)	2.200 (55.88)	1.600 (40.64)	.122 (3.10)
02	17	3	2.470 (62.74)	2.200 (55.88)	1.600 (40.64)	.153 (3.89)
03	23	2	3.070 (77.98)	2.800 (71.12)	2.200 (55.88)	.122 (3.10)
04	23	3	3.070 (77.98)	2.800 (71.12)	2.200 (55.88)	.153 (3.89)
05	29	2	3.670 (93.22)	3.400 (86.36)	2.800 (71.12)	.122 (3.10)
06	29	3	3.670 (93.22)	3.400 (86.36)	2.800 (71.12)	.153 (3.89)
07	35	2	4.270 (108.46)	4.000 (101.6)	3.400 (86.36)	.122 (3.10)
08	35	3	4.270 (108.46)	4.000 (101.6)	3.400 (86.36)	.153 (3.89)
09	41	2	4.870 (123.70)	4.600 (116.84)	4.000 (101.6)	.122 (3.10)
10	41	3	4.870 (123.70)	4.600 (116.84)	4.000 (101.6)	.153 (3.89)
11	47	2	5.470 (138.94)	5.200 (132.08)	4.600 (116.84)	.122 (3.10)
12	47	3	5.470 (138.94)	5.200 (132.08)	4.600 (116.84)	.153 (3.89)

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.

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REQUIREMENTS

Dimensions and configurations: See figure 1, figure 2, and table I.

Mating receptacles: Shall be in accordance with MIL-DTL-21097/4.

Finish: Gold plating in accordance with MIL-DTL-45204, type II, grade C, class 1, over nickel plating in accordance with SAE-AMS-QQ-N-290, class 2, 30 to 150 microinches.

Part or Identifying Number (PIN): M21097/5-(dash number from table I).

Qualification: Not required.

First article testing: First article testing shall be as specified in table II, tests shall be in accordance with MIL-DTL-21097.

Sample size: 8 connectors shall be subjected to the first article testing. All sample units shall be subjected to the inspections in group I. The sample units shall be divided equally into 2 groups of four and subjected to the inspection for their particular groups.

TABLE II. First article tests.

Inspection
Group I (8 samples)
Visual and mechanical inspection
Permeability
Insulation resistance
Dielectric withstanding voltage
Contact resistance
Group II (4 samples)
Temperature cycling
Shock
Bond strength, body assembly
Contact retention
Resistance to soldering heat
Resistance to solvents
Group III (4 samples)
Salt spray (corrosion)
Low-signal level contact resistance
Contact resistance
Bond strength, body assembly

Failures: One or more failures shall be cause for refusal to grant first article approval.

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Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-DTL-21097, this document references the following:

MIL-DTL-21097/4
MIL-DTL-45204
SAE-AMS-QQ-N-290

CONCLUDING MATERIALS

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2022-016)

Review activities:
Army – AR, AT, AV, CR4, MI
Navy – AS, MC, OS, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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